

Beyond Bounds

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Player Manual

[Link here](#)

Legend:

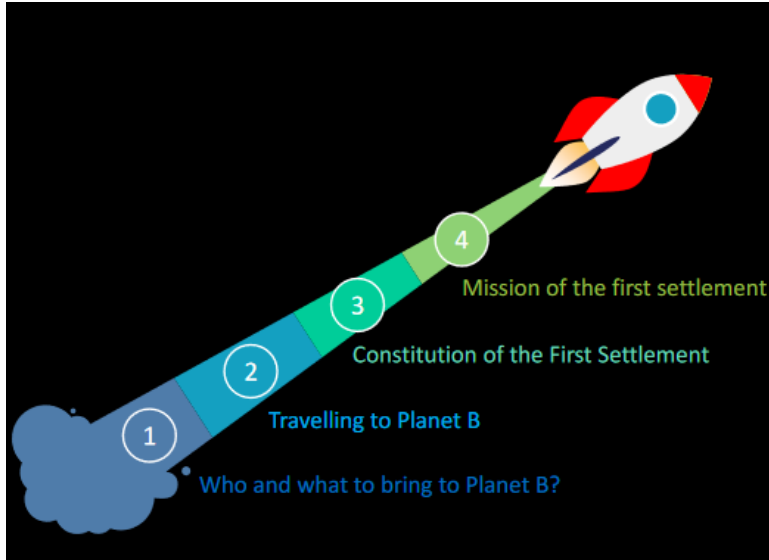
- **Square Parenthesis- []:** If something is in these it means that the values are not final, these can change as the game is still in development

Acronyms:

- **i2p:** Refers to the IdeaSquare Planet cycle.
- **ST:** Supply Tracker
- **RT:** Research Tracker
- **TT:** Time Tracker
- **EC:** Event Cards
- **CC:** Character Cards
- **TC:** Task Cards

What is the game?

Beyond Bounds is a cooperative game where you are part of a team that is sent on a journey based on the IdeaSquare Planet (i2p) cycle that also matches with CERN's DNA. The i2p cycle can be explained through the following image, with the journey back to planet earth being included in the game.



The i2p cycle's goal is to explore and understand how such a migration can be achieved, both in terms of quantitative and qualitative considerations. WE DO NOT have to solve the entire endeavor outright.

The players achieve the following:

- Order of magnitude thinking and doing. (Explored and taught through the gameplay)
- Aiming for seemingly impossible challenges. (Making the game an overall difficult to difficult to successfully complete)
- Questioning assumptions. (Ensuring that the players learn critical thinking skills while playing)
- Reasoned approach to problems. (Achieved through the decision making done by the crew)

The plan is to adapt this into a video game, **WE WILL NOT BE ABLE TO 1:1 RECREATE THIS IN VIDEO GAME FORM. WE DO NOT HAVE THE TIME, AND WILL BE SCOPED DIFFERENTLY ACCORDING TO THE TIME WE HAVE.**

However, the purpose of this is for the basic gameplay and mechanics for a tabletop card game. The game would be based on the tabletop, but not be a 1:1 copy.

How is it played?

Goal of the game and Setup:

The goal of the game is to successfully complete the i2p cycle. This is done by completing each of the 5 phases, and having enough supplies to survive the trip back to earth.

If any of the supplies become less than 0 during phase 2 or 5, the game is over.

The game starts with all supplies on the Supply Tracker (ST) being at [7], with the rest of the tokens, such as the Time Tracker (TT), the research tracker (RT) at 0.

A d10 (a ten sided die) is used to determine skill checks. Reason for this is to statistical math for the rolls should be easy, as a result of a d10 is a 1/10 (10%) chance of happening, so it can be converted to a % easily.

The way the Trackers are used is to place the corresponding token on the number it is supposed to represent. As examples: If you have 5 of each supply, each token is placed on the number 5 on the Supply Tracker (ST). If you have less than 0, you keep the token to the left of zero. If you have 10 or more, you keep the token to the right of the nine.

Player count:

The game is 2-4 players, with each player being a team leader. The four potential positions would be the water team (blue), the food team (green), the health team (red), and the energy team (yellow).

You start the game by having the players select which role(s) they are.

- If 2 players, one player is in charge of two teams
- If 3 players, each player controls a team with the exception of the energy team which becomes a shared responsibility
- (optional/easy mode) If 4 players, each player is a leader of a team.
- (Alternative/less easy mode) If 4 players, the additional fourth player is a helper who can participate in any task.
- ~~(optional) If more players are wanted, they can join in and participate as members. However, these players would not be given leadership roles. Instead work can be delegated by one of the leaders to them, giving them a turn to play. The non-leaders are allowed to participate in the group discussions however, they cannot vote when making a decision.~~

Once the roles are selected amongst the players, the game begins.

Turn order:

To determine turn order (this does not really matter) the starting player must be selected. Whoever is the most well versed in astronomy, is the starting player. Alternatively, this can be done through simply deciding among the group. The next player is then decided for whoever is next in clockwise order. The person next to the starting player, then becomes the new starting player for the next round.

Gameplay loops:

The game mechanics will be explained in further detail in the Game Phases section.

Macro Loop:

1. Players discuss who wants to be what team leader
2. Players decide the character they want to bring on the mission
3. Players go through each phase of the i2p Cycle, and discuss each EC and its potential consequences.
4. Players do their best to survive on Planet B while acquiring as many resources as they can for the journey back

5. They complete their time on Planet B and see if they survived the journey back and beat the game

Micro Loop (happens in Phase 4, the main game):

1. Start a new round
2. Update the Time Tracker (TT). When starting phase 4 set the TT to 9. Every other round, lower TT by 1.
3. A player has a turn
 - a. They choose a character they want to bring along
 - b. Decide what task they wanna do
 - c. Roll to complete task
 - d. Add energy bonuses if they want to
 - e. Place character into Medical Bay if used
4. Next player has their turn until all players have had their turn
5. Research part begins, players discuss where they want to put their research point into
6. Drawn and deal with Event Cards (EC), discuss which choice the crew wants to make
7. Reduce all supplies by 1
8. End of round, repeat step 1 until TT reaches 0
9. Once the TT is 0 you play the final round, and then leave for the journey back to Planet Earth and begin phase 5.

Game Phases:

The game follows the i2p cycle, and has 5 phases. Each phase is concluded with a number of Event Cards (EC). Each EC is a random event that occurs during the journey. These lead to situations occurring that can (either positively, negatively, or indifferently) affect the situation and resources on the journey. ~~After an EC at the end of a phase is pulled, you must lower all supplies by 1, unless stated otherwise, as this is a way to mark that time has passed.~~

Each EC is labeled and color coded for what phase they belong to. With Phase 1 being blue, Phase 2 being light blue, Phase 3 being teal, Phase 4 being green, and Phase 5 being yellow. The back of the card is identical for each phase being labeled, and the front has the prompt explaining the situation and the choices on the top side, with the bottom of the card having the results of the choices made, but written upside down. It is upside down so that whoever draws the card cannot read the results of their answer. (or at least something of the sorts)

(these cards would be similar to what gloom haven have as City/Road cards, examples shown here <https://www.gloomhavencards.com/gh/events>)

The 5 phases are described below:

- **Phase 1- Who and what to bring to Planet B:**
 - In this phase of the game, the leaders need to decide which characters and equipment they want to bring along on their journey.
 - The players must also create their characters in this section, here they choose their stats and ability. The complete process is revealed in the character creation section.
 - (maybe make it that the characters get to name their ship)

- To choose characters the group must decide on [12 out of the 15] available characters. Each character has a power that can be used once per turn, by one of the leaders upon activation. Afterwards the card is placed in the Medical Bay pile and will not be able to be used again, as they become tired, unless restored.
- Examples of character powers can be the following:
 - Energy Expert: When using this power, double the result of the supplies gained if used to gain energy.
 - Extra Assistance: When using this power, a leader is allowed to attempt a task on behalf of another team. So if it is the water leader's turn, he can use a task and apply the results for the medical team.
 - Flash of genius: When using this power, add a +1 to your dice roll.
- Once the characters are decided, separate the [3] non-chosen characters and place them in a separate pile. With the chosen [12], place the deck on top of the Character Cards placement. With the character deck sorted, the equipment must be chosen next.
- The equipment selection is done through deciding how much of each resource the group wants to bring on their journey. (The function of each resource will be described in a later section) The total amount of starting resources is [28]. By default the group will have [7] of each resource, totaling [28] resources. However, the exact number of each resource can be changed according to the group's wishes.
 - As an example, if the players want to have more water to start with, then they can take a point away from food and move it to water. This would make it so that there is 8 water, 6 food, 7 health, and 7 energy.
- The group can use the Supply Tracker (ST) to keep track of the amount of each resource at any given time. And should adjust the ST accordingly to match the supplies chosen.
- Once the supplies are chosen, and ST is updated, the number of ECs are drawn and played out.
- Some examples of ECs for this situation can be the following:
 - The crew did not pack the right amount of hygienic supplies for everyone in the crew. Choose 3 character cards and place them in the Medical Bay pile, as they are sick.
 - The crew forgot to seal the food supplies properly. Reduce the starting food counter on the ST by 1.
 - The crew found that the generators were overclocked. Increase the starting energy counter on the ST by 2.
 - The crew is excited to go on the journey, someone farts really loudly nervously. Nothing happens.
- **Phase 2- Traveling to Planet B:**
 - This phase of the game covers the journey to Planet B.
 - All that happens in the travel sections is that EC's are drawn and played.
 - The number of EC is [3] ~~plus the number of any additional EC that was decided earlier.~~

- The EC that would be in this phase requires the crew to make decisions that affect the journey.
- Example of these ECs are:
 - On the way to Planet B an asteroid field appears in the way. Does the crew vote to go through it or go around it? If you go through it could cause some parts of the ship to be damaged, leading to you losing some supplies (2 food and 1 water) on the way, which makes sections 3-4 harder. However, going around will increase the length of the trip, leading to space time dilation shenanigans. You will still have all of your supplies present, but your time on Planet B is now decreased by 1.
- Once the ECs are completed, and effects are marked down, go to the next phase.
- **Phase 3- Constitution of the First Settlement:**
 - In this phase of the game, you are constituting the first settlement.
 - Here you simply set up the base, as you are starting
 - You then pull [2] ECs and do what comes with them.
 - Once the ECs are completed, the base will be set up, and you can continue to the next phase.
- **Phase 4- Mission of the First Settlement:**
 - In this phase of the game, is where most of the game occurs. Here you are tasked with becoming self-sufficient and surviving on the planet for 5 years.
 - The Time Tracker (TT) is used to mark the passage of time. It starts at 9, and ends at 0. TT goes down by 1 every time a round is completed. Once the TT is set to less than 0 the mission is over, and the crew must embark on the ship and go back to Earth.
 - Time passed is every 6 months (5 years, so 60 months total, so 10 rounds)
 - In this phase, a round is completed after every player has a turn.
 - The turn order is the same as mentioned previously, it begins with the starting player, then goes around the table clockwise. The person next to the starting player, then becomes the new starting player for the next round.
 - A turn starts with the player having to decide which character they want to bring along for the task they want to do.
 - The player then has the choice of either doing their own task team (food, water, medical), or working on an energy task. A task card has 2 prompts on it, one being an easier one with lower rewards, and one being a harder one with higher rewards. If the task is completed successfully, supplies of the respective team are earned.
 - A task requires a roll to complete, the difficulty of the task will vary. A task is beat if the player rolls the same number or higher. If they roll lower, the task is failed and they gain no rewards.
 - At any point before the end of the turn, the player can choose to spend a number of energy supplies to add a +1 per point to their roll.

- Example, If the player rolls a 3 but they need to reach 6, they can spend 3 energy supplies to reach 6, and then look has succeed in the task.
 - Once the task has been completed, if the character power is utilized, place them in the Medical Bay character pile, if not place them back in the pile.
 - If a character is in the Medical Bay they cannot be used again until the required number of rounds (which would say on the card) have passed, or medical supplies have been used. Using a medical supply is the equivalent of 1 round spent resting.
 - Example, if a character was used for a task, and they have 2 days they must spend resting. The players can either wait 2 rounds to play to then be able to play them again, or they can spend 2 medical supplies to immediately have them be available again.
 - Once this has been resolved, the next player in clockwise order does their turn.
 - Once all players have completed their turn, the group gains a Research point. A research point can be added to any section of the Research Tracker. The crew must then decide where they want to put their RT point. The benefits or research will be further expanded on in a later section in the document.
 - During the Research part of this phase, the players are encouraged to openly discuss with each other what they believe is the best option to go forward. If there is a stalemate in the voting, then no points are added to the research tracker, and instead the time is wasted. This is because the crew was not able to successfully cooperate and make a decision.
 - Once the research point has been assigned, an EC is drawn and dealt with.
 - Once the EC is completed, lower all the Supply Tracker (ST) and the Timer Tracker (TT) by 1. This marks the start of a new round and now the next person clockwise starts their turn.
- **Phase 5- Traveling back to Planet Earth:**
 - This phase of the game covers the journey back to Planet Earth, similar to phase 2.
 - All that happens in the travel sections is that EC's are drawn and played.
 - The number of EC is [3] ~~plus the number of any additional EC that was decided earlier.~~
 - The EC that would be in this phase requires the crew to make decisions that affect the journey.
 - Example of these ECs are:
 - Using a similar example to before, on the way back to Planet Earth an asteroid field appears in the way. Does the crew vote to go through it or go around it? If you go through it could cause some parts of the ship to be damaged, leading to you losing some supplies (3 medicine) on the way, However, going around will increase the length of the trip, leading to space time dilation shenanigans. You will still have all of your supplies present, but your time to reach Planet Earth is now increased. So to add .
 - Once the ECs are completed, and effects are marked down, you reach the final part of the game.

- If the crew manage to complete the ECs and you have not run out of any of the supplies, you beat the game and you have completed the i2p cycle.

The Supplies:

What they do:

- Water provides the crew with essential water.
- Food provides the crew with food.
- Medical, helps the crew and characters be physically and mentally healthy.
 - In phase 4, Medical Supplies can be used to make characters return from the Medical Bay pile, and be able to become active again.
- Energy helps the crew perform better, as it can be used as a boost to perform tasks.
 - In phase 4, Energy Supplies can be used to add +1 bonus per point spent to any task check.

What happens when you run out:

- Water, you die as your crew has no water so the mission fails, and the game is over.
- Food, all of the players take a -1 to all their task checks.
- Medical, the time it takes for characters to feel better is increased by 1.
 - Example : You use a character and then they are placed in the Medical Bay. It takes 2 turns for them to recover. The options that you have are that you can use medical supplies to restore a character back to health with 1 medical supply = 1 turn. If you spend your last medical supply on restoring someone, they are restored but everyone else suffers the 0 medical supply penalty. If you have 0 medical supplies, the time it takes for characters to recover is increased by 1. Therefore, if it took you 2 turns to heal a character by default, without using a supply, it takes you 3 turns now before they successfully recover.
- Energy, if in the travel phases (1 and 5) and there is no energy, the ship stops functioning and the voyage fails. Leading to a game over. The lack of energy in phases 2, 3, 4, simply means that the players cannot use energy to boost their rolls.

Researching:

What does it do:

During Phase 4, at the end of everyone's turn, the crew gains a Research point. They can spend this research point on any supply field they want to. The benefits received are universally the same, the difference is which supply they want to improve.

How it works:

Each tier is the same buffs, it's just that the buff changes what supply it is for.

	+1 to roll in that field	Roll 2 d10 and take the higher result for that field	Factor .5 production increase	Roll 3 d10 and take the higher result for that field	Factor 1 production increase
Water					
Food					

Medical					
Energy					

Crew cards(Characters)

Examples:

Below are Ideas, templates, and examples, we can use for Characters/Abilities :

- Energy Expert: When using this power, double the result of the supplies gained if used to gain energy.
- Extra Assistance: When using this power, a leader is allowed to attempt a task on behalf of another team. So if it is the water leader's turn, he can use a task and apply the results for the medical team.
- (there are 5) Flash of genius: When using this power, add a +1 to your dice roll. (there can be 5 of these, one for each team [medical, food, water, energy], and 1 that adds a +1 to anything)
- Specialist Pilot: When using this power add a +3 to any roll that occurs in Phase 2 or Phase 5, as they are an expert in the navigational systems of the ship.
- Captain's speech: When using this power, add a +3 to any roll done for a Moral Check.
- Doctor's Help: When using this power, choose to take a character out of the Medical bay.
- Data Analyst: When taking your turn, choose which tasks you want to follow, then look at the next 3 cards on the top of the deck. You may choose which task card to play, and either place the remaining cards back at the top of the deck or at the bottom.
- Gym Rat: Add a plus +2 to any constitution check made when this character is chosen
- Book Smart: Add a plus +2 to any intelligence check made when this character is chosen
- Mechanical Wisdom: Add a plus +2 to any engineering check made

TTRPG elements:

Make like 6 pre generated characters ? Each character can also have an ability? Maybe make it so that each character has a unique ability, but the players choose their stats.

Each character should have a name, a description, 4 attributes, a role?

Maybe?

- Intelligence, (Int)
 - How well you know things, how quickly you can do research ?
- Constitution, (Con)
 - You physical constitution, how well u can withstand the rough environment of the planet, how well u can do physical tasks
- Morale, (Mor)
 - How well they can mental constitution, how well you can support the crew in the mission
- Engineering, (Eng)

- How well you can repair things, and work with machines.

Make it so that the tasks you have to do can be related to one of these attributes?

Probably have it that the pregenerated characters are completed and try not to implement character creation, as it would be a half assed attempt squeezing RPG elements into a game that does not need it. Similar to how Gloomhaven does it.

Heroes (Player Pregens):

Every character has to have a total of 0 in each stat, and you start with 0 in everything by default.

Stat Spreads Example:

The Default Character: 0 Int, 0 Con, 0 Mor, 0 Eng

Optimized: -1 Int, 0 Con, 1 Mor, 0 Eng

Min-Max: -2 Int, 2 Con, 0 Mor, 0 Eng

Two players cannot have the same ability, an ability can be activated at any time where applicable.

Abilities examples (these need to be stronger than character card abilities):

- Bastion of Hope: You are an inspiring leader to the crew aboard your ship. Once per round you can give any player other than yourself a +3 on a morale check, or a +2 on any other check.
- Resident Doctor: Once per round you may temporarily remove a character from the medical bay without expending any medical supplies, placing them back at once they have been utilized.
- Energy Efficient: Every time you complete a task, you gain an additional energy supply.
- Peak Physique: Every time you need to make a constitution check you add a +3 to that roll.
- Engineering GOAT: Once per round, you can add a +4 to any engineering check.
- Scrumptious Chef: Anytime you complete a food or water task, you gain an additional supply of that kind.

Event Cards

Below are Ideas, templates, and examples, we can use for event cards :

Phase 1 - Who and what to bring to Planet B:

~~In this section there would be EC that would affect the starting conditions of the mission. Events that could occur relating to not packing supplies correctly or certain members of the team you picked doing something.~~

- ~~● The crew did not pack the right amount of hygienic supplies for everyone in the crew. Choose 2 character cards and place them in the Medical Bay pile, as they are sick.~~

- ~~The crew forgot to seal the food supplies properly. Reduce the starting food counter on the ST by 1.~~
- ~~The crew found that the generators were overlocked. Increase the starting energy counter on the ST by 2.~~
- The crew is excited to go on the journey, someone jitters nervously really loudly.
- ~~Someone in the crew was sick and managed to get other crew members infected. Roll a 8+ On a success some of the crew members feel better, Place 2 Characters in the Medical Bay. On a failure, Place 5 characters in the Medical Bay,~~
- ~~The crew forgot to purify some of the water containers aboard, and they have gotten unsafe for consumption. Reduce the starting water counter on the ST by 2.~~
- ~~The crew did a final check on all supplies that were loaded on board. Everything seems to be fully operational and functioning. Increase Mood Meter by 1~~
- ~~The crew noticed that some of the equipment in the energy facility was faulty. Roll an Engineer Check 3+ on a success, you fix the equipment, nothing bad happens. On a failure, Reduce the starting energy counter on the ST by 2.~~

Phase 2- Traveling to Planet B:

This section would be short and would focus on events that could occur while you are on the ship on your .

- ~~During your travels, an asteroid field appears, blocking your path. Does the crew vote to go through it or go around it? If you go through it could cause some parts of the ship to be damaged. However, going around will increase the length of the trip, leading to space time dilation shenanigans.~~
 - ~~If you chose to navigate through the asteroids, you must Roll 7+.~~
 - ~~**On Success:** nothing happens.~~
 - ~~**On Failure:** it leads to you losing some supplies on your trip. (2 food and 1 water)~~
 - ~~If you choose to go around. You will still have all of your supplies present, but your time has now decreased by 1.~~
- A lot of the people in the crew seem to be nervous, fidgeting. What do you do? Do you try to increase mood via sea shanty, although it might backfire as people might not get into it? Or simply let them deal with it on their own. (2 min)
 - (A) Roll a 7+. You start to sing a sea shanty, the rest of the crew joins along. Spirits amongst the crew are uplifted, increasing the starting medical supply on the ST by 1. On a success Increase the Mood Meter by 1, on a fail the free cringes, and nothing else happens.
 - (B) You let the crew deal with their own nervousness, *fail*, decrease the Mood Meter by 1.
- A set of resting pods malfunctions, due to this many people are no longer able to sleep comfortably. Do you have faith in them? Roll a 8+ on a success nothing happens. In an

attempt to try to fix the resting pods, which you know is difficult, or do you let the crew handle it?

- (A) You try to fix it, make an engineering check, 5+, to see if you can repair the resting pods. On success, you succeed and *Increase the Mood Meter by 1*. On a failure you make it worse and the crew tries to fix your mess. *Reduce the number of medical supplies on the tracker by 4*.
- (B) You let the crew repair the resting pods. They sleep uncomfortably for a short while, but they manage to repair it relatively quickly. *Reduce the number of medical supplies on the tracker by 2. Lower the Mood Meter by 1*.
- A fight breaks out within the crew and you manage to break it up. Someone in the crowd tells you that the fight was messy, 4 people were involved, and they don't know who started it. Do you want to resolve the fight peacefully, or do you blame one person and only punish them? (A) You try to resolve things peacefully, make a 7+ Morale check. If you succeed in the check, the people involved calm down and go back to work. *Nothing happens*. If you fail, everyone involved is dissatisfied and constantly blames the others for this mess. It escalates, and another smaller fight breaks out. Everyone is injured. Place the top 4 characters on the Character deck into the medical bay. Decrease the Mood Meter by 2
 - (B) You decide to minimize the potential damage that the best way to approach this is to blame one person and end it here. You blame them and tell them to take time off. Place the top card in the character deck and place them in the medical bay. Decrease the Mood Meter by 1
- The navigation system has an unusual freeze, it suggests a new path towards Planet B, saying that the current course would take longer than planned. In order to fully reboot it you need to wait a week. Do you follow the new path or Reboot the system and try to follow the path after that?
 - (A) You follow the new path, and in reality the navigation system was malfunctioning, you now have less time on planet B. Reduce the Timer Tracker by 1 (start at 8). Reduce Mood Meter by 1
 - (B) You follow the original course that you have been following, mid way through following this path, the timer corrects itself and goes back to being the original time. *Increase the Mood Meter by 1*.
- You and your crew encounter an asteroid belt blocking your path. (A) Will you decide to spend additional time traveling around the asteroid belt or (B) will you take a chance and guide your crew through the unknown?

The team has 7 min to make their decision.

- (A) You choose to go through. You now have to roll for possible damage in the spaceship.
 - If the team gets an odd number, then you get hit.
Mood remains the same, but you have gotten out of trajectory and need to spend two more years traveling. Decrease all resources by 2.
 - If the team gets an even number, then you make it through.
Everyone is thrilled. Mood increases by 2.
- B) If you choose to go around, the mood increases by 1, but you have to spend one more year traveling to correct the trajectory. Decrease all resources by 1.
- Note for GM: If the team does not make the choice under 7min then they are forced to go through. Roll for damage:
 - If the team gets an odd number, then you get hit.
Mood decreases by 1, but you have gotten out of trajectory and need to spend two more years traveling. Decrease all resources by 2.
 - If the team gets an even number, then you make it through. Lucky you! Nothing changes.

Phase 3- Constitution of the First Settlement:

These events take place when you land on Planet B and are settling the first base. Here would occur events like optimizing the base's setup or finding new things when initially landing on the planet.

- ❖ Upon landing on the new planet. You need to place special sensors outside to measure the levels of radiation, which is necessary to ensure the safety of the whole crew throughout the mission. Two people volunteered to go out on this dangerous expedition to place the first sensor. A 50 year old and an inexperienced but ambitious 20 year old. Who do you choose to send, knowing that the older volunteer will need medical attention afterwards while the younger volunteer could make it without harm but there is also the chance for much darker consequences. You have seven minutes to make a decision.
 - (A) You have decided to send the 50 year old on this journey
 - 1- If the mood meter is neutral, nothing happens.
 - 2- if the mood meter is positive, the 50 year old returns back from the journey, he will need medical attention but all went well. *Reduce the medical supply by 2.*
 - 3- if the mood meter is negative, the crew begins to protest against the decision as their lower mood has led to unrest. They begin to ignore the rationing rules and start to eat as they please. Eventually, the 50 year old returns back from the journey, he needs medical attention. *Reduce the mood by 2 points, the medical supply by 2, as well as reducing the food and water supplies by 1.*
 - (B) You have decided to send the younger person on this journey, she is swift and ambitious but also a little overconfident.

- **When the team comes to a decision within 7 minutes after drawing the card,**
 - Roll a Die;
 - ◆ **Numbers 1,3,5,7,9:** The Volunteer did not lock the helmet properly. A crew medic notices it once the volunteer return from placing the sensor, she is placed into quarantine and supplied with medicine, but unfortunately does not make it. **Decrease Medicine by 1, Decrease Mood by 1, Decrease Energy by 1.**
 - ◆ **Numbers 0,2,4,6,8:** The is briefed by other, more experienced crew members, and manages to plant the sensor without harm or injury. The crew celebrates, **increasing their mood by 2.**
- **After 7 minutes of the team discussing without coming to a decision,** the young volunteer cannot contain themselves anymore, grabs and locks in the helmet and enters the evacuation bay.
 - Roll a Die;
 - ◆ **Numbers 1,3,5,7,9:** The Volunteer did not lock the helmet properly. A crew medic notices it once the volunteer returns from placing the sensor, she is placed into quarantine and supplied with medicine, but unfortunately does not make it.
 - **Decrease Medicine by 2**
 - **Decrease Mood by 2**
 - **Decrease Energy by 2**
 - ◆ **Numbers 0,2,4,6,8:** The Volunteer manages to plant the sensor and return without harm or injuries. **Nothing gets lost.**
- Upon landing and settling down on the base in Planet B you notice that the base facility doesn't have any windows. *Nothing happens.*
- The crew is unloading everything and everyone is helping. You notice that a fire starts in the supply room and you only have time to save one set of supplies. Which one do you save? Water, Food, Medical, Energy. Roll a 6+ Physical. (1 min)
 - On a success, *Reduce the supply tracker by 1 for all the supplies you have not chosen.*
 - On a fail, *you fumble the resource, leading you to lose 2 for the chosen supplies, and 1 supply for everything else. Regardless of roll, Reduce Mood Meter by 1.*
- You are setting up the initial base and you notice that your sensors seem to briefly detect a larger water source in a different area than planned. Do you take the time to relocate the water facility, forcing you to change plans, or do you keep it in the same place and save time and resources? (You have 3 mins discuss)

- (A) If you relocate the water facility you spend the time to properly adjust and place the facility, it turns out the ping is correct and the reserve of water is larger than the previous one. *Reduce the Timer Tracker by 1, and increase the starter water supply by 4. Increase Mood Meter by 2.*
 - (B) You decide to stick to the information you know and decide to not delay the constitution of the settlement. *Nothing changes*
- You notice that the positioning of the energy base can be improved, as the solar panels on the base could be positioned better. Do you try to calculate and change the location of the solar panel and attempt to improve it or do you leave it as is saving time and resources?
 - (A) You try to improve the positioning of the solar panels, make a 6+ Intelligence check to optimize the location of the solar panel. On success you move the solar panel to a better location, increasing energy supply by 1. On a fail you think you've placed it on a better spot, however, it seems that it was not the case. Reduce the energy supply by 2.
 - (B) You leave it as is. *Nothing happens*

Phase 4- Mission of the First Settlement:

This section is the biggest part of the game, and occurs while surviving on Planet B. Anything can happen here it can affect any part of the game.

- You have a breakthrough in your research, *you gain an additional +1 research point. Increase mood meter by 1.*
- Your medical bay has a breakthrough. Place every character in the medical bay back into the crew deck. Increase Mood Meter by 1.
- Electrical Failure occurs throughout the facilities. (1 min) Roll for Engineering 7+, On a save Increase Mood Meter by 1, on a fail, For the next round you cannot spend any energy supplies to increase your rolls. Decrease Mood Meter by 1
- The monthly rock-paper-scissors tournament ends in a draw. Everyone celebrates the successful event. Increase Mood Meter by 1.
- Famine. Your food supply gravely suffers as something happened to it. (2 min) Roll for Intelligence 5+, *on a success you save most of the crops but you lose 2 food supplies, on a fail, You lose 5 of your food supply. Decrease Mood Meter by 2*
 - *Can probably do it so that Famine has 1 set path of EC, leads to other ECs linearly.*
- A disease spreads throughout the base, and a couple of your crew gets sick. (1 min) *Take the bottom 2 cards from your crew deck and place them in the medical bay. Decrease Mood Meter by 1*
- You have the chance to go on an expedition to set up a tower, potentially leading to getting extra research done. You have the option to go on this expedition for a research purpose mission where you need to choose some of the crew mates to come with you.

(A)- You will be required to complete 3 tasks successfully, or fail the expedition. (B)- You save your time and resources. (You have 4 mins what do you chose?)

- ~~○ If you want to commit to it, Choose two characters to help you on your journey, and take a minimum 1 of each supply with you, so you reduce the supply trackers by 1. You then will need to make 3 checks of differing kinds, with flavor text explaining the situation you are in. Upon Succeeding everything you receive a research point as an award. If you fail one or more things nothing happens.~~
- ~~○ If you do not commit you save time and resources.~~

(A) You want to commit to the expedition, Choose two characters to help you on your journey, and take 2 of the water, food, and medical supplies with you from the total supply.

You then will need to make a 6+ Constitution check, as you are traversing the planet, followed by a 3+ Morale check, to keep your crew's mental state throughout the journey, and a 5+ Engineering check to successfully build the tower.

Upon Succeeding everything you receive a research point as an award. Increase Mood Meter by 3

If you fail one or more things nothing happens. Decrease Mood Meter by 2

(B) You do not commit, you save time and resources.

- ~~X environmental condition happens, and the player needs to choose the correct response and deal with it properly, or suffer consequences. Ex (that is not scientifically accurate)~~ A sandstorm occurs, a generator is blown out. Do you choose to leave the generator be during the sandstorm, and potentially have no power for a while or try to fix it during the storm. (2min)
 - (A) You choose to repair it before it is afterwards. Choose a character you wish to bring with you. Make a 5+ constitution check, to see if you can endure the sandstorm. Then, you must make a 6+ engineering check to see if you can successfully repair the generators, on success Increase Mood Meter by 2.
 - If you fail, your character gets sent to the medical bay and you lose 2 medical supplies. Decrease Mood Meter by 1.
 - (B) You do not attempt to repair the generator, and reduce your total supply to that number. Decrease Mood Meter by 2

Phase 5- Traveling back to Planet Earth:

This section would be short and would focus on events that could occur while you are on your ship. (probably can use the same events as Phase 2, just reworded)

- ❖ A medic crew member has fallen ill during the journey and is in critical condition. (A) Will you sacrifice resources to help them now or (B) will you leave them in their condition until you reach the planet. (you have 3 mins discuss)

➤ Roll a 5+ Int to save this crew member.

- **On success:** You realize that the crew member is very dehydrated. You have successfully saved the crew member. When they wake up, they provide your team with additional resources for medical supplies. (-1 to water supply)(+2 to medicine supply)(increase mood meter by 2)
- **On Failure:** You have failed to save this crew member. As a result, your medical supplies have decreased. Also, Because of this, your crew's mood has decreased.(mood meter -1) (-2 to medical supply)

Task Cards

Each task starts in one of the difficulties and can be chosen to move to the next tier of difficulty for increased rewards.

Easy Tasks:

Anything that requires a 3+ roll (roll a 3,4,5,6,7,8,9,10 to succeed)

- Regular maintenance of equipment is essential, you must upkeep the water purifiers. Make a 3+ engineering roll, on success gain 1 water supply, on failure nothing happens. (to make it harder make a 5+ roll and gain 2 water supplies)
- Your food crew seems to be bored farming the food. Make a 4+ Morale roll, on success gain 2 food supply, on a failure nothing happens. (to make it harder make a 6+ roll and gain 3 food supply)
- You are tasked with completing a basic health check up. Make a 2+ Constitution roll, on success gain 1 medical supply. On a failure nothing happens. You cannot make this task harder.
- You need to replace the batteries for a device. Make a 3+ Intelligence roll, on success gain 1 energy, on a failure nothing happens. (to make it harder make a 5+ roll and gain 2 energy)

Medium Tasks:

Anything that requires a 5+ roll (roll a 5,6,7,8,9,10 to succeed)

A generator seems to be malfunctioning, you must go and fix it. Roll a 5+ on an engineering check. On success gain 2 energy, on a failure nothing happens.

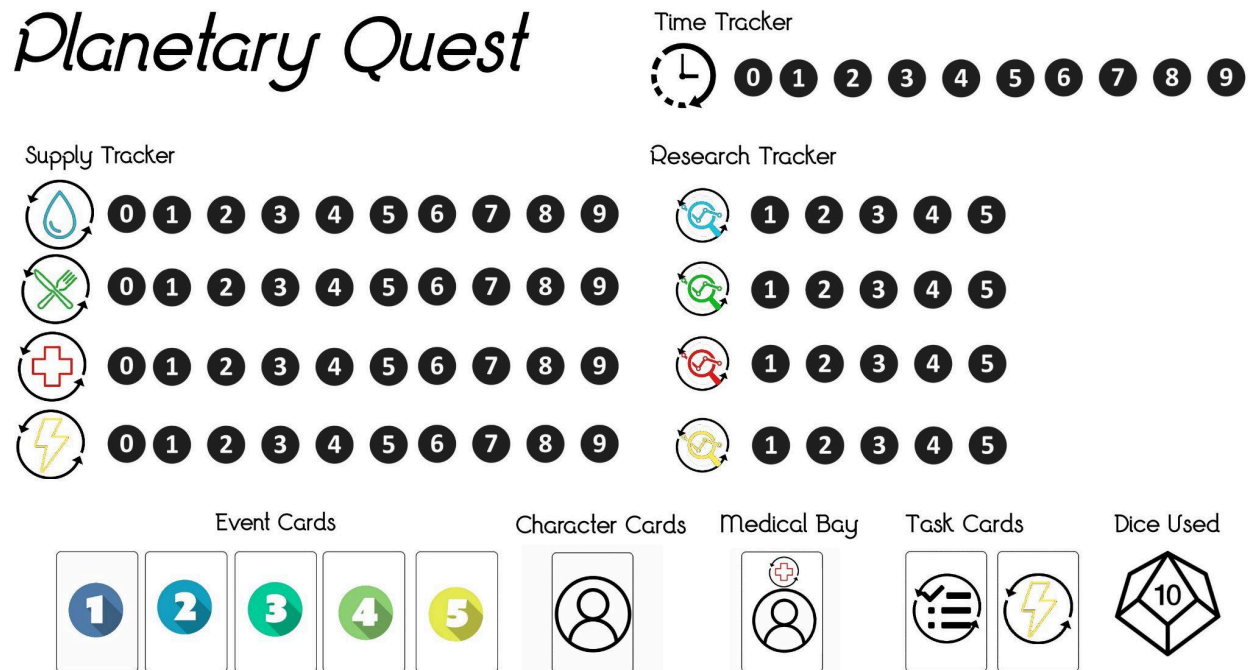
An injured crew member is starting to get depressed. Help them back up to feeling better. Roll a 6+

Difficult Tasks:

Anything that requires a 7+ roll (roll a 7,8,9,10 to succeed)

Visualization

Below is a visualization for how the various board pieces would look like.



How does Planetary Quest achieve OoMT?

Order of magnitude thinking and doing. (Explored and taught through the gameplay)

An order of magnitude is an exponential change of plus or minus 1 in the value of a quantity or unit. (<https://www.techtarget.com/whatis/definition/order-of-magnitude>).

Factor 1 increase of something means that it was increased by x10. Factor 2 increase of something means that it was increased by x100.

OoMT is achieved in this game through some of the Event Cards and through the Research done. Through exploring the situations and gaining an understanding of what is happening in the game, the players expose themselves to OoMT. By playing the game they will have to tackle

and understand the value of OoMT, and what it really means in context. Through the use of the Research abilities, the players can see how big a factor 1 or 2 difference is.

How can the skills applied can be used for Famine?

The skills applied though playing this game will be achieved in this game through some of the Event Cards and through the cooperative critical thinking that the players do in order to beat the game.

Another way the skills learned through the game can be applied to real world situations would be through the ECs. Through ECs and their explorations of scenarios that would be found in a Famine, or similar to them. The EC would be specifically designed to teach the skills necessary to resolving a famine. As an example, an event card happens where the food or water supply is decreased significantly, it would lead the players to think of ways to solve their own in-game famine.

Assets used:

(for visualization)

<https://static.vecteezy.com/system/resources/previews/009/317/654/original/set-of-environmental-ecology-icon-logo-illustration-eco-friendly-pack-solar-energy-wind-turbine-nuclear-water-resource-and-etc-symbol-template-for-graphic-and-web-design-collection-vector.jpg> (Tasks)

<https://i.pinimg.com/originals/1a/0a/bc/1a0abcf1ab67c1c9fcd45512747685ab.jpg> (Card Outline)

https://t4.ftcdn.net/jpg/01/18/11/53/360_F_118115323_OqUQTAjSpBBijwqHNFKtTQJz4YFhkYfz.jpg (numbers)

<https://cdn3.vectorstock.com/i/1000x1000/18/92/line-energy-hazard-symbol-with-arrows-around-vector-15211892.jpg> (Energy)

<https://media.istockphoto.com/id/1327954795/vector/vector-set-of-restaurant-line-icons-contains-icons-menu-serving-food-chef-wine-list-cutlery.jpg?s=612x612&w=0&k=20&c=7IkSoJrKfdPZ9qUUSCNSJVEkdqXriidZWb3BzHOox00=> (food)

https://t4.ftcdn.net/jpg/05/26/75/07/360_F_526750715_kaliAfn43a1s7IRprLdOK0vcAjyExwcM.jpg (water)

https://media.istockphoto.com/id/1220379891/vector/medical-assistance-line-icon-medical-cross-or-plus-care-outline-style-pictogram-on-white.jpg?s=612x612&w=0&k=20&c=_ojEzZGx_ZF8xNbQUUp7IHNS_g0EIQbwJGSBAMcaYk= (Health)

https://t3.ftcdn.net/jpg/03/61/19/98/360_F_361199867_PAx7XWLqwoSZdcYAzRx3yQtmXmtZIt p.jpg (numbers resources)

https://media.istockphoto.com/id/1179573547/vector/user-profile-login-or-access-authentication-icon-vector-illustration-image.jpg?s=612x612&w=0&k=20&c=rnHRXIPL7flUaUWxyWJ0epW17X56fg_RKX2CluHpSJ4= (Person)

https://static.vecteezy.com/system/resources/previews/000/633/173/non_2x/clock-icon-symbol-sign-vector.jpg (clock)

<https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcRC9mSmMYmYILUh989CI3xwz5oIRSQES5HtKLHbMWZDMA&s> (d10)

<https://www.creativefabrica.com/wp-content/uploads/2021/04/05/Market-Research-Icon-Graphics-10404267-1-1-580x386.jpg> (Research)

<https://icons.veryicon.com/png/o/miscellaneous/standard/task-32.png> (Task)

Extra Ideas:

I think that in this section we can add any recommendations/ideas we can have for the game. As an example, maybe a way of making the game more than 4 players.

Or even adding a mechanic that shows OoMT more, like having different sized physical board game pieces increasing in size for when u make a certain production increase by Factor 1 or 2

Optional difficulty challenges can be added. Add rules or scenarios that change the way the game is playing. Ex. making it that you start with way less supplies, at the beginning of the game.

We can try to make this game run in Tabletop Simulator.

Different biomes?

WE CAN ADD A VARIABLE FOR ENVIRONMENTAL FACTORS (ADD THINGS WITH PHYSICS END) like gravity, solar radiation, access to different terrain, (is there water is there atmosphere) we can do this and scale it for different planets. (we currently focus on mars-like and moon-like planets) Add environmental variables that can occur there.

We meeting with cern ppl so we can ask them how to make it more scientifically accurate

Karenarose & Edrex- Notes from meeting 4/25 for progress:

- Need: Writers of Event Cards (Minimum 5 for each of the four phases). Each event card has an inciting incident, a player choice, and corresponding consequences dependent on player choice. Impact can vary, implications of choice can be subtle/opaque.
- Need: Visual and text explainers for OoMT that help players respond to each corresponding event card
- Environmental -- gravity, solar radiation, water, atmosphere -- weighting of values for resource management (Mars and the moon are the most likely targets.)
- The Narrative Team takes ownership of Event Cards writing.
- Karenarose will work with Marco RE: choices
- Joey will ownership of visuals; need guidelines/criteria for designing visual assets from Narrative/Mechanics Team
- Visual Team should move forward with Mars-like terrain right away

CERN meeting:

Knowledge generation

Expansion, such as for different planets

We can also make the game focus more on a specific irl problem,

Implement character creation and TTRPG elements?

Have a group of ppl be in each team, and structure it like classes in dnd. Not each team does their independent thing.

Edrex notes @ CERN meeting:

Nature of Game play—

Constant group discussion

Decision Making with high stakes

What are the driving motivations? How do we win?

Team formation? Quick Communication?

Character choice — choose team, choose research

Establishing characteristics, boundary conditions

Increase complexity

Should we character as part of experience (D&D, or Epic MetaHuman)

The existing pilots rarely have teams represented by single characters. People should be represented to encourage a feeling of belonging.

What do other creatures look like? What do they represent? Power Differential, Conflicting Physiological Processes

Confront explorers with pitfalls of colonization thinking — “tension”

Geneva Experience:

Play Minimum Viable Product to iterate, and identify emerging principles
For student team to experience themselves, MVP: Tabletop Assets, Tabletop simulator

Feedback from playtest:

- Make the TT more obvious that it is for phase 4 and make TT 5 not 10
- Write [player manual](#)
- Change the research so that it is half of a factor production increase
- Pregen characters are the move, make sure that they are balanced
- Improve the way the board looks, try to add the whole game
- Change armstrong to be total 0 (buff armstrong)
- Add more variety (the whole game being EC is kinda meh)
- Make it that the 1,2,3 have bigger permanent impacts on phase 4, like that you do an oopsie and your energy is now a max of 7.
- Decisions on EC should be clear on the consequences, like having each decision explained what happens on it. So A makes you lose X and gain Y, B nothing happens, C you gain Y and lose Z .
-

Feedback from playtest @CERN:

Work on sanity,

There is a line

Leadership styles

Universally a leader is someone who has more of a say into the different decisions that are being made for the EC. Depending on the style the different leaders will have different powers, and the turn order of who is leader would change.

We can either make it that the players have to think of this on their own, they can choose one of the ones we write down, or we can draw a random card and that is what they are required to do.

Transformational:

- Transformational leadership is a leadership style that emphasizes change and transformation. Leaders who adopt this approach strive to inspire their followers to achieve more than they ever thought possible by tapping into their potential.
- In this style whoever wants to step up as leader for the turn can take the leadership token and make decisions.

- There is a token that is passed around between the players, the people can volunteer to be leader, and whoever steps up first gets the position.

Delegative (Laissez-Faire)

- Participative leaders encourage group members to participate, but retain the final say in the decision-making process.
- The leaders are chosen at the beginning of the game by whoever rolls the highest
- Whoever is the leader is not required to participate in the discussion, however, they retain the final say in what decision they are doing.
- It is the job of the other players to convince the leader for what decision is better

Authoritative (Autocratic)

- Authoritarian leaders, also known as autocratic leaders, provide clear expectations for what needs to be done, when it should be done, and how it should be done.
- This style of leadership is strongly focused on both command by the leader and control of the followers. There is also a clear division between the leader and the members.
- Authoritarian leaders make decisions independently, with little or no input from the rest of the group.
- In this version, whoever rolls the highest is chosen as the leader.
- They have full control of how they want to operate the mission and cooperate with the crew

Participative (Democratic)

- Democratic leaders offer guidance to group members, but they also participate in the group and allow input from other group members.
- Participative leaders encourage group members to participate, but retain the final say in the decision-making process.
- This style of leadership you would have a token/prop to pass around, in clockwise order.
- Everyone is allowed to participate and work together to choose the best solution

<https://www.verywellmind.com/leadership-styles-2795312>

<https://www.imd.org/reflections/leadership->